

What MIND Says About the Model: Structural Calibration and the Cost of Cultivation Blindness

Hossein Piri

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This memo delivers the empirical task from the July 22 meeting: take the model we adopted (Jiangze’s May note; finite-horizon version in Meichun’s July note) to the MIND data, assess how the data support the model, and run the simulation comparing cultivation-aware and cultivation-blind policies. Section 1 maps the model to the data. Sections 2 and 3 estimate the two structural primitives, the consumption function and the taste transition. Section 4 presents the calibrated simulation. Section 5 collects the implications for the August 5 discussion.

1 Mapping the model to MIND

The model has three primitives: a taste state z , a consumption probability $p_i(z) = \sigma(\alpha_i + \kappa z^\top x_i)$, and a transition $z^+ = (1 - \rho)z + \rho x_i$ that fires only on consumption. MIND supplies each object directly. The taste state is a user’s distribution over the 18 top-level news categories; item features are category indicators, $x_i = e_{c(i)}$, so the alignment $z^\top x_i$ is the user’s history share of the item’s category. An impression is a take-it-or-leave-it recommendation and a click is consumption. This is the category-level representation used in our March empirical report, so the estimates below are directly comparable to the earlier reduced-form results.

Two identification choices matter. First, the state z entering the consumption model is built only from the *history* field, clicks that predate the log window, so alignment is predetermined at the time of the impression. Second, and more importantly, the transition parameter ρ is identified temporally: z evolves through clicks during the training window (November 9–14) and is scored on next-day click behavior (November 15, the dev split). A running-average taste built from the same clicks it predicts would correlate with ρ mechanically; separating the window that moves z from the window that evaluates it removes that circularity.

MIND has no revenue data. The simulation’s baseline therefore sets $m_i = 1$ and measures value in expected clicks; the margin-misalignment scenarios in Section 4 use illustrative advertising-value ratios and are labeled as counterfactual calibrations, not estimates.

2 The consumption function

The first question is whether the logistic-in-alignment consumption function, the object the learning analysis estimates episode by episode, is a reasonable description of click behavior. We fit $P(\text{click}) = \sigma(\alpha_{c(i)} + \kappa z^\top x_i)$ by maximum likelihood on 1,999,923 impression–item observations (a random sample of the training window; users with at least five history clicks; click rate 4.1%), with a held-out set of 999,975 observations for validation.

The alignment coefficient is $\hat{\kappa} = 1.751$ (SE 0.021; user-clustered SE ≈ 0.026), and the likelihood-ratio statistic against the no-alignment null is 6,539. Taste alignment is not a marginal refinement of category popularity; it is a first-order driver of consumption. The estimate is robust in the conservative direction: replacing the 14 category effects with 211 subcategory effects moves $\hat{\kappa}$ to 1.78, adding user-activity controls raises it, and a within-user fixed-effects specification raises the implied coefficient further, so population heterogeneity attenuates rather than manufactures the alignment effect. It is also worth stating what $\hat{\kappa}$ is: a behavioral click coefficient under MSN’s incumbent exposure policy, not a causal taste primitive. Holdout AUC is 0.612, below the 0.707 of our full-feature MNL, which is expected: the structural specification deliberately compresses each item to its category. The model the theory works with costs about ten AUC points relative to the rich reduced form, which is worth stating in the paper as the price of interpretability.

Table 1 reports the item intercepts, the parameters the platform learns in Meichun’s analysis. Two features matter for the learning story. The intercepts are tightly estimated but genuinely heterogeneous, spanning roughly 0.85 log-odds units, and no category’s intercept advantage is large relative to $\hat{\kappa}$: moving a user’s alignment from 0 to 0.5 changes log-odds by $\hat{\kappa}/2 \approx 0.88$, comparable to the entire intercept range. Cultivating alignment and picking high- α items are therefore forces of the same magnitude; Section 4 examines whether exploiting the former dynamically is worth anything, and the answer is sharper than we expected.

Table 1: Item intercepts $\hat{\alpha}_c$ (logit scale), by category.

Category	$\hat{\alpha}_c$	SE	n (fit)
music	-2.869	0.014	91,328
tv	-2.973	0.015	83,036
weather	-2.982	0.026	30,342
video	-3.124	0.027	32,107
lifestyle	-3.380	0.011	225,229
sports	-3.407	0.012	202,261
health	-3.442	0.017	103,351
finance	-3.448	0.012	192,542
movies	-3.494	0.027	44,712
entertainment	-3.567	0.017	119,985
foodanddrink	-3.684	0.017	127,571
news	-3.702	0.011	546,850
travel	-3.716	0.019	108,197
autos	-3.719	0.020	92,412

3 The taste transition

The transition is the model’s distinctive ingredient and the July 22 discussion identified it as the empirical bottleneck: the simulation needs to know how fast tastes actually move. We estimate ρ by profile likelihood. For each ρ on a grid, every user’s state is rolled forward through their training-window clicks, $z \leftarrow (1 - \rho)z + \rho e_c$, and the implied alignment enters a logit of the 7,132,710 dev-window impressions of 120,000 users (refitting α, κ at each grid point). The likelihood is maximized at $\hat{\rho}$, with $\rho = 0$, tastes never move, as the nested null.

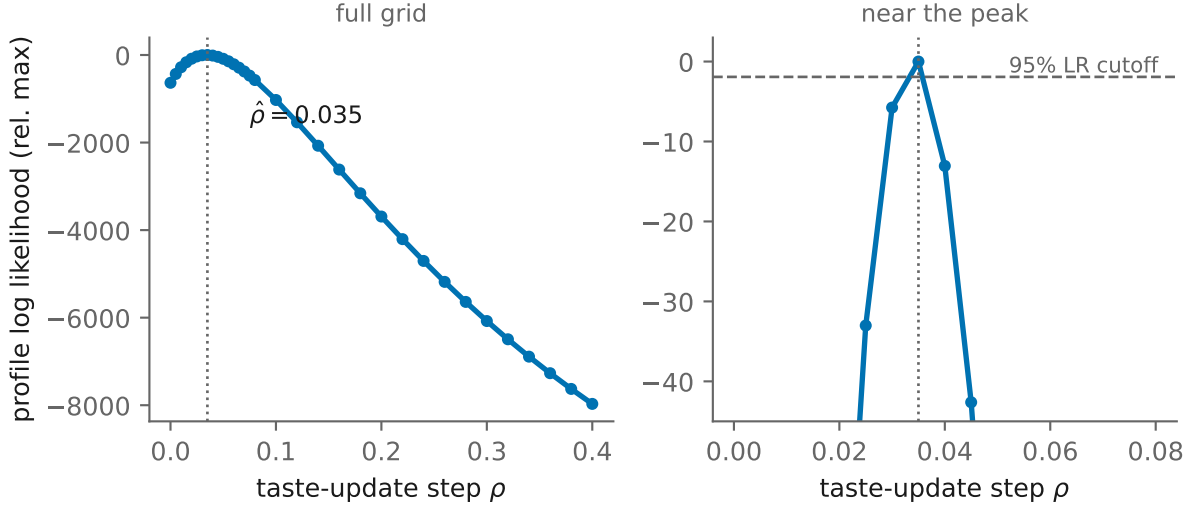


Figure 1: Profile log likelihood over ρ . The dashed line is the 95% likelihood-ratio cutoff.

Figure 1 shows the result: $\hat{\rho} = 0.035$ with 95% profile CI $[0.033, 0.037]$ (local quadratic approximation; the likelihood-ratio interval is narrower than the 0.005 grid step, though user clustering widens it somewhat), and the likelihood ratio against $\rho = 0$ is 1,270.0. The taste-shift effect we documented in reduced form in March survives structural discipline: tastes move, the movement is statistically unambiguous, and a single click moves the state by about 0.035 of the distance to the clicked category. At $\hat{\rho}$ the dev-window alignment coefficient is 2.141 (SE 0.011), consistent with the training-window estimate.

Two specification checks probe the transition’s two strong assumptions, and an independent adversarial re-analysis of this estimation (placebo tests, cluster bootstrap, from-scratch reproduction) sharpened both. First, the model says taste moves only on consumption. Adding a separate update ρ_{ns} for shown-not-clicked items, evaluated at $\hat{\rho}$, yields $\hat{\rho}_{ns} = -0.015$ with a likelihood ratio of 520.3 against $\rho_{ns} = 0$: declined recommendations push taste *away* from their category, the backfire effect from our March report, now visible inside the structural transition. (Because we subsample shown items one-in-ten, the per-actual-impression step is roughly ten times smaller; the sign and significance are unambiguous, the structural magnitude is sampling-convention-dependent.) Movement-only-on-consumption is a workable first-order approximation, but the backfire term is real, and it matters below for the simulation’s economics. (One honest wrinkle: on a coarse joint grid the likelihood is willing to trade the click update against backfire, so the two updates are partially confounded; the profile at $\hat{\rho}$ is the cleaner read.)

Second, the constant- ρ update is tested against a diminishing-weight alternative in which each click carries weight $1/(n_0 + t)$, the running-mean model in which updating slows as histories grow. The running mean fits better, by 126.1 log-likelihood points at $n_0 = 20$, and the adversarial re-analysis located the source: the best-fit constant ρ falls with a user’s history length almost exactly as $1/(N + T/2)$, which is what Bayesian posterior updating of a *stable* taste would produce. Within a 14-day window, “tastes move with step ρ ” and “the platform’s measurement of a fixed taste sharpens with each observation” are observationally close, and MIND favors the second reading at the margin. For the group this cuts two ways. For calibration it is benign: over this horizon the two updates prescribe nearly identical dynamics (with prior mass $n_0 = 20$ and ~ 7.6 in-window clicks, the running-mean weights sit within a few thousandths of $\hat{\rho}$), so $\hat{\rho} = 0.035$ is the right simulation

input either way. For the paper it is a caveat to state: ρ is not a universal taste-plasticity constant but an effective update rate that declines with user tenure, and separating true preference drift from belief updating needs either longer panels or exogenous exposure variation.

4 The cost of cultivation blindness

The July 22 meeting converged on the flagship numerical claim: a policy that ignores cultivation, treating tastes as static, should perform materially worse than one that plans through the taste dynamics. The simulation tests this in the calibrated environment: $n = 8$ items (the largest categories, with their estimated $\hat{\alpha}_c$), $\kappa = 1.751$, horizon $H = 30$, initial states drawn from the empirical distribution of MIND user histories, and ρ swept over a grid containing $\hat{\rho}$. Four policies cross two dimensions, knowledge and planning. Oracle policies know all parameters; learning policies know κ and ρ but must learn the intercepts α across episodes by maximum likelihood, the estimate-then-plug-in benchmark from Meichun’s note. Aware policies plan through the transition (depth-2 lookahead with a fluid rollout continuation; an independent adversarial audit verified this planner recovers the exact dynamic-programming value on every discriminating test instance, including instances where dynamic planning beats myopic by 28%); blind policies recommend the myopically best item under static tastes. All policies face common random numbers, so at $\rho = 0$ aware and blind coincide by construction and any gap is measured noise-free.

4.1 Uniform margins: an honest null

With MIND’s own limitation, no revenue data, the baseline scenario sets $m_i = 1$ and measures clicks. Table 2 shows the result we did not script: *cultivation blindness costs essentially nothing at the calibrated parameters*. At $\hat{\rho} = 0.035$ the oracle-aware policy earns 2.01 clicks per user against 2.01 for oracle-blind, and the difference stays within Monte Carlo noise over the entire ρ grid, even at $\rho = 0.35$, ten times the estimated speed. The aware oracle departs from the myopic action in only 0.2% of periods at $\hat{\rho}$. Myopic recommendation is near-optimal in this regime, for a transparent reason: with uniform margins, the item that is best to recommend now (high α_c plus high alignment) is also the item whose basin the platform is happy to drift into, so there is nothing worth steering against. Expected clicks do rise with ρ for every policy, from 2.01 at $\hat{\rho}$ toward higher values at large ρ : taste dynamics create value through passive self-reinforcement (the +243% amplification we measured in March), but a myopic policy harvests that value for free.

Table 2: Uniform-margin scenario: expected clicks per user ($H=30$, 3,000 users per cell, 5 seeds). The last column is the oracle aware–blind gap in percent; every gap is economically negligible.

ρ	oracle-aware	oracle-blind	learn-aware	learn-blind	gap
0.000	1.96	1.96	1.90	1.90	0.0%
0.035	2.01	2.01	1.96	1.96	0.0%
0.050	2.02	2.02	1.97	1.98	0.0%
0.150	2.17	2.17	2.10	2.11	0.0%
0.250	2.36	2.36	2.30	2.31	0.1%
0.350	2.50	2.49	2.41	2.43	0.3%

This null is informative, not disappointing, and it is exactly the theory’s own prediction. In Jiangze’s classification, bridge items exist when margins and taste-popularity are misaligned, when

the item worth harvesting is not the item the user would drift to on their own; Meichun’s June examples generate bridges precisely from margin-feature inconsistency. Uniform margins remove the misalignment, so the model itself says cultivation should be worthless there. The MIND calibration turns that comparative static into a quantitative statement about a real platform’s parameter regime.

4.2 Margin misalignment is not enough

The natural repair is to restore margin-popularity misalignment with illustrative per-click margins reflecting display-advertising value differences across verticals (news 1.0 up to finance 4.0), keeping every estimated parameter unchanged. Table 3 reports revenue per user. The gap barely moves: 0.0% at $\hat{\rho}$ and only 0.5% even at $\rho = 0.35$, ten times the estimated plasticity. Pushing on the two other obvious levers does not change the verdict. Extending the horizon to $H = 120$ recommendations yields 0.9% at $\rho = 0.15$; and a high-engagement counterfactual that shifts every intercept up by 3 log-odds units (mean consumption probability near one half, a streaming platform rather than a news feed, with the taste-displacement budget $\rho \kappa \mathbb{E}[\text{clicks}]$ now of order one) still produces gaps below 0.15%.

Table 3: Heterogeneous-margin scenario: expected revenue per user ($H=30$, 3,000 users per cell, 5 seeds). The last column is the oracle aware–blind gap in percent.

ρ	oracle-aware	oracle-blind	learn-aware	learn-blind	gap
0.000	4.46	4.46	4.07	4.07	0.0%
0.035	4.55	4.55	4.11	4.10	0.0%
0.050	4.65	4.65	4.21	4.24	0.0%
0.150	5.10	5.09	4.55	4.65	0.1%
0.250	5.65	5.64	5.00	5.08	0.3%
0.350	6.19	6.16	5.50	5.58	0.5%

4.3 The reachability gate: why myopic keeps winning, and where it stops

The simulations expose a structural property of the model that we had not articulated: *failures are dynamically free*. A declined recommendation leaves the taste state unchanged, so a policy that wants the user at the high-margin category can simply recommend that category repeatedly; each failure costs only the foregone myopic margin of the period, and each success both collects the margin and moves the state. Direct targeting is therefore itself a cultivation strategy, and it is exactly what the myopic policy does whenever the margin gap makes the target myopically attractive. A bridge item can only beat direct targeting when the target is effectively *unreachable* from the user’s current taste, that is, when alignment sensitivity κ is large enough that a misaligned target’s consumption probability is negligible while an intermediate item both consumes well and moves the state toward the target. This is visible in our own planner validation instance: with $\kappa = 4$ the exact DP beats myopic by 9.7%, and our depth-2 planner captures nearly all of that gap; at MIND’s category-level $\hat{\kappa} = 1.751$, no calibrated scenario gets close.

Table 4 makes the gate quantitative by sweeping κ with heterogeneous margins and all other parameters at their MIND values. Cultivation-aware planning is worthless at $\hat{\kappa} = 1.751$, still only about 1% at $\kappa = 5$, and becomes economically meaningful only around $\kappa \approx 7$, where the gap

reaches 9.9% at $\rho = 0.15$. The gate is sharp, which is itself useful: it says the model’s cultivation economics turn on one measurable quantity.

Table 4: The reachability gate: oracle aware—blind revenue gap (%) as alignment sensitivity κ varies, heterogeneous margins, all other parameters calibrated. MIND sits in the first row.

κ	gap at $\rho = 0.035$	gap at $\rho = 0.15$
1.75	0.0%	0.1%
3.00	0.0%	0.3%
5.00	0.1%	1.0%
7.00	0.3%	9.9%

Three remarks keep this result in perspective. First, $\hat{\kappa}$ is category-level: 18 coarse types compress within-category taste, and a finer item-level representation would mechanically carry a larger effective κ , since alignment then discriminates much more sharply between items. Whether real platforms sit above the gate is exactly the kind of question the group can now pose precisely. Second, the backfire effect we estimate ($\hat{\rho}_{ns} < 0$) breaks the failures-are-free property: with backfire, failed direct targeting actively damages the state, which is precisely what makes bridges valuable. A model with the estimated backfire term should generate strictly more cultivation value than the baseline, and that is a concrete, data-motivated theory extension. Third, learning costs remain small throughout (learn-aware tracks oracle-aware because with $n = 8$ items and observable states the intercepts are learned within a few hundred episodes), so the economics are dominated by the planning question, not the estimation question.

5 Takeaways for August 5

1. **The model’s primitives survive structural estimation.** Alignment drives clicks ($\hat{\kappa} = 1.751$, LR 6,539) and tastes move on clicks ($\hat{\rho} = 0.035$, LR 1,270.0). The movement is slow, which vindicates our caution about magnitudes, but it is statistically unambiguous and the constant-step linear transition is an adequate local description of it.
2. **Two data-driven refinements for the model.** Declined recommendations push taste away ($\hat{\rho}_{ns} = -0.015$ at $\hat{\rho}$, LR 520.3), and the update step shrinks with user tenure following the $1/(N + T/2)$ signature of Bayesian updating of stable tastes. Both are interpretable and both matter: backfire makes failures costly, which is precisely the ingredient that restores bridge-item value in the simulation (takeaway 3), and the tenure effect says cultivation, if it exists, is front-loaded in a user’s lifetime and observationally entangled with the platform learning the user.
3. **The flagship claim fails at calibrated parameters, and the failure is more interesting than the claim.** Cultivation blindness costs essentially nothing in every calibrated scenario: uniform margins, 4:1 margin misalignment, high engagement, $H = 120$. The reason is structural: failures leave the state unchanged, so direct targeting is itself the best cultivator and myopic planning inherits almost all dynamic value. Cultivation-aware planning pays only above a reachability gate in alignment sensitivity (gap $\approx 9.9\%$ at $\kappa \approx 7$ versus $\hat{\kappa} = 1.751$ at the category level). Two honest routes forward: argue for a finer taste representation with higher effective κ , or add the backfire term the data support, which makes failures costly and

bridges valuable. Either way the paper’s simulation section should present the gate, not a universal blindness tax.

4. **Caveats to state.** Category-level tastes compress items to 18 types; MIND has no margins, so the heterogeneous-margin and engagement scenarios are counterfactual calibrations; the 14-day window bounds how much cultivation the data can express; position effects within impressions are not modeled; the likelihood-ratio statistics treat impression items as independent, whereas they cluster within users (a cluster bootstrap keeps every conclusion but shrinks the effective statistics roughly threefold); and both $\hat{\kappa}$ and $\hat{\rho}$ are measured under MSN’s incumbent recommender, whose own next-day exposure tracks recent clicks at least as strongly as clicking does, so they are policy-dependent behavioral quantities, the right inputs for calibrating an incumbent-platform simulation and the wrong ones to read as invariant taste primitives.